

Oscar Antonio Juarez Jimenez
AI / Machine Learning Engineer | LLM, RAG & Distributed Systems

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AI Engineer with 2+ years of experience building and shipping end-to-end data, machine learning, and large language model systems in banking and enterprise environments. Expertise in PySpark ETL pipelines on AWS, containerized microservices with Docker and Kubernetes, and CI/CD-driven deployments. Proven track record translating ambiguous business problems into measurable outcomes, shipping AI tools from prototype to production and driving real adoption, with LLM agents and conversational assistants in daily use by business teams.

EXPERIENCE

Data Engineer | Bluetab, an IBM Company

Jul 2025 – Present

Client: BBVA | Mexico City, MX

- Cut end-to-end data preparation time by 40% for analytics and machine learning teams by designing and deploying production-grade PySpark ETL pipelines that integrate 10+ source systems into the bank's core data platform.
- Reduced Spark job cost and data scanned volume by 22%, measured as cluster time per job, by restructuring datasets and tables on AWS S3 with strategic partitioning and enforced schema governance.
- Unblocked 10+ machine learning use cases in the structural risk division by translating feature requirements into production-grade data products adopted directly by modeling teams and risk analysts.

AI Engineer, Innovation & Development | Grupo Salinas

Feb – Jul 2025

Internship | Mexico City, MX

- Halved manual handling time for administrative workflows, reducing average resolution time per request by 50%, by deploying LLM-based agents and intelligent chatbots built with LangChain and OpenAI APIs.
- Shortened prototype-to-validation cycles by delivering interactive prototypes and dynamic landing pages within a Design Thinking process, enabling business units to test AI concepts before committing engineering resources.

PROJECTS

3D Object Segmentation in Indoor Scenes using NeRF and Diffusion Models

Jun 2024 – Aug 2025

B.Eng. Thesis, IPN-ESCOM | PyTorch, NeRF, Stable Diffusion

- Improved segmentation accuracy by 30% over the U-Net baseline, measured by mean Intersection over Union (mIoU) across 200 annotated indoor scenes, by combining classical machine learning classifiers with a U-Net architecture implemented in PyTorch.
- Achieved photorealistic 3D scene editing by integrating NeRF spatial reconstruction with diffusion-based texture synthesis from Stable Diffusion, and fully documented the architecture and evaluation protocol for reproducibility.

Retrieval-Augmented Generation (RAG) Chatbot for Mexico's Official Gazette (DOF)

Jan – Jun 2024

Personal Project | LangChain, LangServe, Pinecone, Docker

- Cut legal document search time from approximately 1 hour to 90 seconds per query—a 97% improvement—by building a Retrieval-Augmented Generation system that answers and summarizes natural-language questions over Mexico's Official Gazette (Diario Oficial de la Federación).
- Shipped the system as a containerized Model Context Protocol (MCP) server, leveraging LangChain and LangServe for orchestration, Pinecone as the vector database backend, and Docker for reproducible builds and seamless deployments.

EDUCATION

B.Eng. in Artificial Intelligence | Instituto Politécnico Nacional (IPN) — ESCOM

Jul 2025

GPA: 9.5/10 (equivalent to 3.8/4.0)

Relevant coursework: Machine Learning, Neural Networks and Deep Learning (foundations and advanced), Natural Language Technologies, Digital Image Processing, Parallel Computing, Bio-inspired Algorithms

SKILLS

Machine Learning & AI | Deep Learning, Neural Networks, Convolutional Neural Networks (CNN), U-Net, Computer Vision, Natural Language Processing (NLP), Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Evaluation, Error Analysis, PyTorch, TensorFlow, scikit-learn, LangChain, LangServe, Pinecone, Stable Diffusion

Cloud, MLOps & Infrastructure | Amazon Web Services (AWS) — S3, Elastic MapReduce (EMR), AWS Glue, Identity and Access Management (IAM); Google Cloud Platform (GCP); Microsoft Azure; Docker; Kubernetes; CI/CD; Git/GitHub; MLflow; Model Deployment and Monitoring

Data Engineering | Apache Spark, PySpark, Distributed Computing, Extract-Transform-Load (ETL) Design, SQL, Query Optimization, Data Modeling, Schema and Partition Governance, Vector Databases

Programming Languages | Python, Java, SQL, R, Scala, C/C++, JavaScript/TypeScript, PHP, Go, CUDA, Bash

CERTIFICATIONS

AWS Certified Cloud Practitioner · AWS Certified AI Practitioner · Anthropic Academy — Full Course Track · Google Cloud Computing Foundations · Google Cloud Generative AI Skill Badges · Microsoft Learn AI Skills Challenge (Azure) · SAFe Scrum Master (Scaled Agile, Inc.)

LANGUAGES

Spanish — Native | English — Professional Working Proficiency (C1)